Day 10 Notes

Advanced Python Concepts- Enumerate

Note: In python, index starts from 0

1. enumerate()

enumerate() adds a counter to an iterable and returns it as an enumerate object.

Syntax:

```
enumerate(iterable, start=0)
```

Example:

```
fruits = ["apple", "banana", "cherry"]

for index, value in enumerate(fruits):
    print(index, value)
```

Output:

```
0 apple
1 banana
2 cherry
```

Why use it?

- When you need both index and value while looping.
- Cleaner than using range(len(list)).

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2. delattr()

delattr() deletes an attribute from an object.

Syntax:

```
delattr(object, attribute_name)
```

Example:

```
class Student:
    name = "Sriya"

delattr(Student, "name")
```

---

3. isinstance()

isinstance() checks whether an object belongs to a specific class.

Syntax:

```
isinstance(object, class)
```

Example:

```
x = 10
print(isinstance(x, int))  # True
```

---

4. issubclass()

issubclass() checks whether a class is a subclass of another class.

Syntax:

issubclass(child_class, parent_class)

Example:

```
class Animal:
    pass

class Dog(Animal):
    pass

print(issubclass(Dog, Animal))  # True
```

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5. Miscellaneous Built-in Functions

I. globals()

Returns a dictionary of the current global symbol table.

```
x = 10
print(globals())
```

Used to access global variables dynamically.

II. locals()

Returns a dictionary of the current local symbol table.

```
def test():
    a = 5
    print(locals())

test()
```

III. callable()

Checks whether an object can be called like a function.

```
def greet():
    pass

print(callable(greet))   # True
print(callable(10))      # False
```

IV. eval()

Evaluates a string as a Python expression.

```
x = "5 + 3"
print(eval(x))  # 8
```

Be careful: eval() can be dangerous if used with user input.

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6. Exceptional Handling

Exception handling prevents program crashes and handles runtime errors gracefully.

I. try

Code that may cause an error is placed inside the try block.

II. except

Handles the error if it occurs.

III. finally

Always executes, whether an error occurs or not.

Example:

```
try:
    num = int(input("Enter a number: "))
    result = 10 / num
except ZeroDivisionError:
    print("Cannot divide by zero!")
except ValueError:
    print("Invalid input!")
finally:
    print("Execution completed.")
```

---

7. del

Deletes objects or variables.

```
x = 10
del x
```

After deletion, accessing x will cause an error.

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8. Garbage Collection (gc)

Python automatically manages memory using **Garbage Collection (GC)**.

Garbage collection removes unused objects to free memory.

I. gc.collect()

Manually triggers garbage collection.

```
import gc
gc.collect()
```

II. gc

The gc module provides functions to control garbage collection.

```
import gc
print(gc.get_count())
```

Assignment 3:
Testing Enumerate, delattr, isubclass, isinstance

enu1.py

Users > sriyamk > Desktop > Infosys Springboard > My_repo > assignments > assignment_3 > enu1.py

```python
a = ["S", "for", "Sriya"]
# Iterating list using enumerate to get both index and element
for i, name in enumerate(a):
    print(f"Index {i}: {name}")

# Converting to a list of tuples
print(list(enumerate(a)))

fruits = ["apple", "banana", "cherry"]

for index, value in enumerate(fruits):
    print(index, value)


#isinstance
x = 10
print(isinstance(x, str)) #10 is not a string, returns False
y= "Hello"
print(isinstance(y,str)) #y is a strigng
```

```python

#issubclass
class animal:
    pass
class dog(animal):
    pass
print(issubclass(dog, animal))

#testing enumerate
K = "bobby", "sally", "josh"
for i, name in enumerate(K):
    print(i,name)


# delattr
class student:
    name = "Sriya"
    age = 20
delattr(student,"name")

#purposefully trying to access the deleted attribute to show that it has been removed
try:
    print(student.name)
except AttributeError:
    print("The attribute 'name' has been deleted.")
```

Output:

Problems Output Debug Console Terminal Ports Python

```
sriyamk@Sriyas-MacBook-Pro ~ % /usr/bin/python3 "/Users/sriyamk/Desktop/Infosys Springboard/My_repo/assignments/assignment_3/enu1.py"
Index 0: S
Index 1: for
Index 2: Sriya
[(0, 'S'), (1, 'for'), (2, 'Sriya')]
0 apple
1 banana
2 cherry
False
True
True
0 bobby
1 sally
2 josh
The attribute 'name' has been deleted.
sriyamk@Sriyas-MacBook-Pro ~ %
```

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